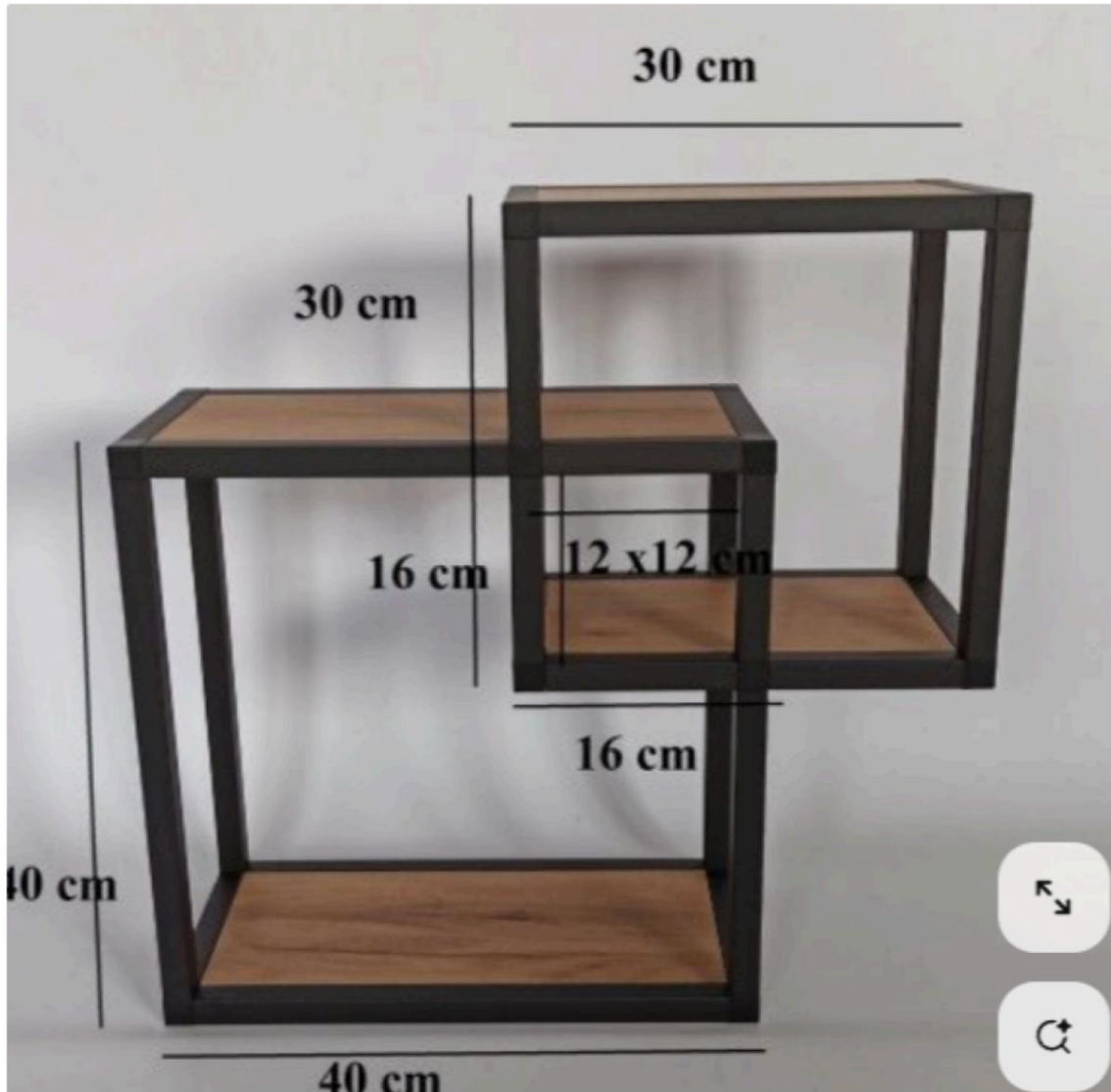


+100 Furniture Projects (part 1 - 15 pieces of furniture)

Proyecto 1: Mueble de repisas escalonadas



1. Description of the furniture

Decorative piece of furniture with a tiered design, ideal for displaying plants, books, or decorative items. Its minimalist matte black metal frame and wooden shelves make it perfect for modern and industrial interiors.

2. General measures

- Total height: 74 cm

- **Total length:** 40 cm
- **Depth:** 30 cm

3. List of materials (metal profiles)

Material	Dimensions s	Thickness (sheet)	Length	Quantity
Metalon square	20x20 mm Sheet Metal 18		40 cm	4 pieces
Metalon square	20x20 mm Sheet Metal 18		30 cm	6 pieces
Metalon square	20x20 mm Sheet Metal 18		16 cm	6 pieces
Metalon square	20x20 mm Sheet Metal 18		12 cm	2 pieces

Additional material:

- 3 MDF or wood boards measuring 30x30 cm (suggested thickness: 1.5 cm)
- Matte black anti-corrosion paint
- Screws for fixing shelves

4. Necessary tools

- Welding machine (MIG or inverter)
 - Angle grinder with cutting disc
 - Drill and bits for metal/wood
 - Tape measure and set square
 - Safety glasses and gloves
 - Hand sander or sandpaper
 - Paint or aerosol gun
-

5. Cutting details

- **40 cm:** 4 cuts (lower and upper base of the large module)
 - **30 cm:** 6 cuts (horizontal and upper sides of medium modules)
 - **16 cm:** 6 cuts (vertical sides of intermediate module and support)
 - **12 cm:** 2 cuts (structure of the small central decorative support)
-

6. Difficulty level

Low – Ideal for beginners with basic knowledge of welding and tools.

7. Estimated manufacturing time

2 to 3 hours (depending on experience and available tools)

8. Assembly instructions

1. **Cut all profiles** according to the indicated measurements.
 2. **Assemble the lower module:** weld 2 40 cm profiles with 2 30 cm profiles forming a rectangle; add 40cm vertical legs.
 3. **Build the second level** (intermediate module): join 30 cm profiles with 16 cm profiles and fix them to the right side of the first module.
 4. **Assemble the central decorative support** with 12 cm and 16 cm profiles, forming a partial cube.
 5. **Roughen the welds** with the grinder to improve the finish.
 6. **Paint the entire structure** with matte black anti-corrosion paint.
 7. **Screw the wooden shelves** onto the already dry metal structures.
-

9. Additional tips

- Use a **magnetic square** when welding to maintain perfect angles.
- Sand and clean the metal well before painting for better adhesion.
- You can **modify the dimensions** of the shelves to adapt them to other spaces.

Proyecto 2: Librero industrial escalonado



Project 2: Stepped industrial bookcase

1. Description of the furniture

Modern metal frame bookcase with wooden shelves arranged in a tiered. Ideal for displaying books, plants or decorative items. Its open and vertical makes it functional for homes, offices or studios with an industrial style.

2. General measures

- **Total height:** 193 cm
- **Total width:** 80 cm
- **Estimated shelf depth:** 30 cm (suggested)

3. List of materials (metal profiles)

Material	Dimension is	Thickness (sheet)	Length (cm)	Approx. quantity.
Metalon square	25x25 mm Sheet 18		193 cm	2 pieces (vertical sides)
Metalon square	25x25 mm Sheet 18		80 cm	5 pieces (spars main horizontals)
Metalon square	25x25 mm Sheet 18		30 cm	12 pieces (inner profiles in stepped Z)
Metalon square	25x25 mm Sheet 18		20 cm	6 pieces (vertical reinforcements internal)

Additional material:

- 6 MDF or wood shelves measuring approximately 30x30 cm.
- Matte black anti-corrosive paint

- Screws or rivets to secure the shelves
-

4. Necessary tools

- MIG or electric welding machine
 - Angle grinder (cutting and grinding disc)
 - Drill and bits for metal and wood
 - Tape measure, level and set square
 - Safety glasses and gloves
 - Sander or sandpaper
 - Brush or spray paint
-

5. Cutting details

- **193 cm:** 2 pieces (vertical sides)
 - **80 cm:** 5 pieces (top, bottom and horizontal crossbars)
 - **30 cm:** 12 pieces (horizontal step-type intermediate profiles)
 - **20 cm:** 6 pieces (internal vertical supports to divide the shelves)
-

6. Difficulty level

Medium – Requires precision in cuts and squaring to ensure shelves fit leveled.

7. Estimated manufacturing time

5 to 6 hours (with a break for painting and assembly)

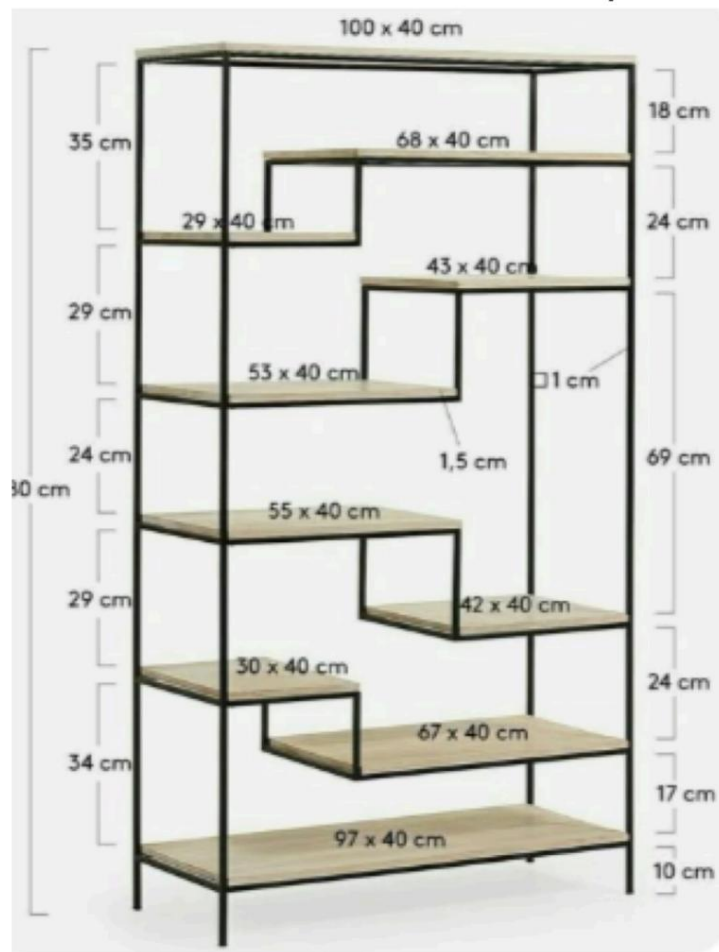
8. Assembly instructions

1. **Cut all metal profiles** according to the indicated measurements.
 2. **Assemble the main outer frame:** join 2 193 cm side panels with 2 80 cm profiles (top and bottom).
 3. **Add the intermediate horizontal profiles** where the shelves will go.
 4. **Form the stepped Z-type structures**, welding the 30 cm and 20 cm profiles as internal support for each shelf.
 5. **Reinforce the joints with additional weld points.**
 6. **Sand all joints**, clean with degreaser and paint the entire structure in black.
 7. **Secure the wooden shelves** with screws from the bottom or hidden brackets.
-

9. Additional tips

- Be sure to square weld each section to maintain right angles.
- You can adjust the number or size of the shelves if you want more storage space.
- If you will be using it against a wall, you can add **rear fixing brackets** for greater stability. security.

Proyecto 3: Librero industrial con repisas en zigzag



Project 3: Industrial bookcase with zigzag shelves

1. Description of the furniture

This industrial-style vertical bookcase features an original zigzag shelving arrangement, perfect for adding a modern and dynamic touch to living rooms, offices, or studios. The combination of black metal and light wood gives it strength and elegance, allowing you to organize books, decorative objects, and more.

2. General measures

- Total height: 180 cm
- Total width: 100 cm

- **Standard depth:** 40 cm
- **Shelf thickness:** 1.5 cm
- **Heights between levels:** Varied (see sections)

3. List of materials (metal profiles)

Material	Dimension is	Thickness (sheet metal)	Length (cm)	Approx. quantity.
Metalon square	25x25 mm	Sheet Metal 18	180 cm	2 pieces (vertical sides)
Square metalon	25x25 mm	Sheet 18	100 cm	2 pieces (top and bottom front)
Square metalon	25x25 mm	Sheet Metal 18	40 cm	14 pieces (depths, supports and interior reinforcements)
Metalon square	25x25 mm	Sheet 18	Varied (between 29–69 cm)	12 pieces approx. (internal horizontal supports according to shelf)

Additional material:

- 7 MDF or pine wood shelves of different dimensions (see cutting section)
- Matte black anti-corrosion paint
- Invisible internal screws or brackets

4. Necessary tools

- MIG or inverter welder
- Angle grinder (cutting and grinding disc)
- Drill and bits for metal and wood
- Tape measure, level, set square

- Safety glasses and gloves
- Hand sander or sandpaper
- Spray paint or brush

5. Cutting details (shelves)

Shelf	Measurement (cm)
Shelf superior	100 x 40
Shelf 2	68 x 40
Shelf 3	43 x 40
Shelf 4	53 x 40
Shelf 5	55 x 40
Shelf 6	42 x 40
Shelf 7	30 x 40
Shelf 8	67 x 40
Lower shelf	97 x 40

6. Difficulty level

Medium – Requires good planning of cuts, squaring and progressive assembly in height.

7. Estimated manufacturing time

6 to 8 hours, including assembly by sections, cleaning, painting, and shelf mounting.

8. Assembly instructions

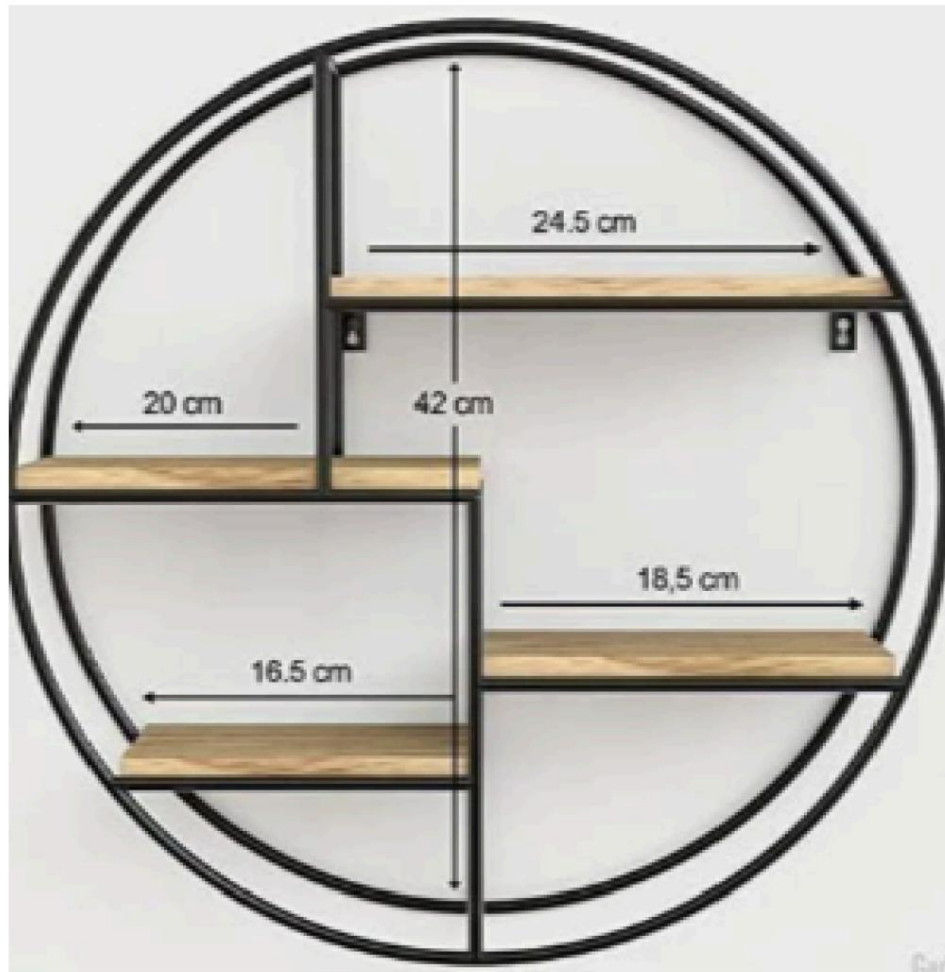
1. Cut all metal profiles to the required measurements.

2. **Assemble the outer structure of the bookcase:** 180 cm vertical sides and 100 cm horizontal crossbars (top and bottom).
 3. **Fix the depth reinforcements (40 cm)** that will serve as a base for the shelves.
 4. **Weld the zigzag supports** according to the positions corresponding to each level of shelf.
 5. **Reinforce critical points** with additional welding and level visually.
 6. **Sand and clean the structure well**, then apply black anti-corrosion paint.
 7. **Place and secure the wooden shelves** with screws or invisible fixings.
-

9. Additional tips

- Check that the **joints are exactly square** to avoid visual imbalance.
- You can **adapt the dimensions of the shelves** according to your space needs.
- Add leveling pads to the legs if the floor where you place it is not even.
- If it is against a wall, consider adding a **rear mounting bracket**.

Proyecto 4: Estante circular con repisas metálicas



Project 4: Circular shelf with metal shelves

1. Description of the furniture

Wall shelf with a circular metal frame and modern design. It features asymmetrically arranged wooden shelves, ideal for displaying small decorative objects, plants, or candles. Its minimalist and elegant aesthetic makes it a centerpiece for living rooms, hallways, or bedrooms.

2. General measures

- **Outer ring diameter:** 60 cm (estimated by image proportions)
- **Approximate depth:** 10 cm (to accommodate shelves and structure)

- **Height from base to top:** 60 cm
- **Height of the central crossbar:** 42 cm

3. Bill of materials (metal structure)

Material	Dimensions	Thickness (sheet)	Length	Amount
Round bar solid	Ø 10 mm (steel)	—	188 cm approx.	2 curved sections in circle
Metalon square	15x15 mm Sheet 18		42 cm	1 piece (vertical) central)
Metalon square	15x15 mm Sheet 18		20 cm	1 piece
Metalon square	15x15 mm Sheet 18		18.5 cm	1 piece
Metalon square	15x15 mm Sheet 18		16.5 cm	1 piece
Metalon square	15x15 mm Sheet 18		24.5 cm	1 piece

Additional material:

- 4 MDF or pine wood shelves (measurements according to the following section)
- Screws or bracket-type fasteners (for fixing wood to metal)
- Matte black paint for metal

4. Necessary tools

- MIG or electric welder
- Bending machine or roller to form the ring (it can also be made in a workshop)
- Angle grinder

- Drill with metal/wood bits
 - Sandpaper and spray paint
 - Level, tape measure and set square
-

5. Cutting details (wooden shelves)

Shelf	Measurement (cm)
Top right	24.5 x 10
Top left	20 x 10
Bottom left	16.5 x 10
Bottom right	18.5 x 10

6. Difficulty level

Intermediate/Advanced – Requires precise hoop bending and wall mounting.

7. Estimated manufacturing time

4 to 5 hours (less time if the ring is bent in an external workshop)

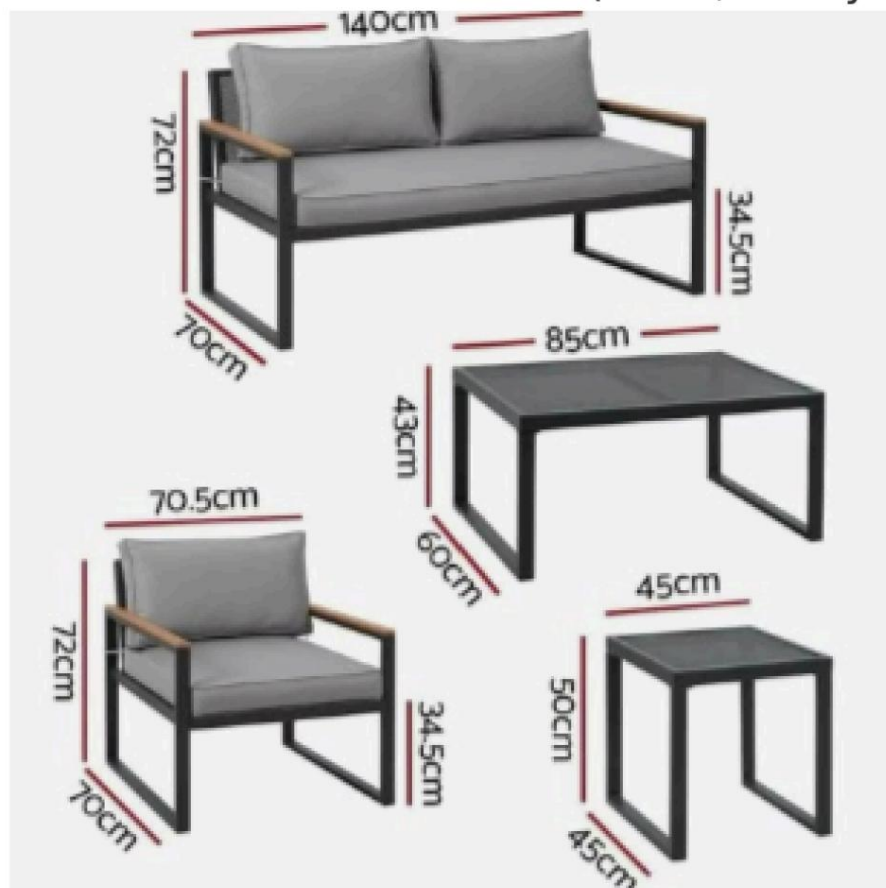
8. Assembly instructions

1. **Bend two round bars** to form the double hoop of 60 cm diameter.
2. **Weld the** 42 cm central vertical crossbar right on the axis of the ring.
3. **Add the horizontal crossbars** according to the measurements of each shelf.
4. **Sand the joints**, clean with degreaser and paint black.
5. **Place the wooden shelves** on the crossbars and secure them with screws.
small from below or hidden supports.
6. **Install metal hangers** or rear holes for wall mounting.

9. Additional tips

- You can use **circular hollow bars** if you want to lighten the weight of the shelf.
- For rim precision, it is ideal to use a **workshop with industrial rollers**.
- If the shelf will be heavily loaded, use **butterfly anchors** for greater support. the wall.

Proyecto 5: Set de sala industrial (sillón, sofá y mesas)



Project 5: Industrial living room set (sofa, armchair, and tables)

1. Description of the furniture

This industrial set consists of a two-seater sofa, a single armchair, a coffee table, and a side table. The design combines a metal frame with wood and cushions, making it ideal for interiors or covered terraces. Its minimalist industrial style makes it an excellent decorative and functional option.

2. General measures

Couch:

- Width: 140 cm
- Depth: 70 cm
- Total height: 72 cm
- Seat height: 34.5 cm

Single armchair:

- Width: 70.5 cm
- Depth: 70 cm
- Total height: 72 cm
- Seat height: 34.5 cm

Coffee table:

- Width: 85 cm
- Depth: 60 cm
- Height: 43 cm

Side table:

- Width: 45 cm
- Depth: 45 cm
- Height: 50 cm

3. Bill of materials (metal structure)

Part	Suggested profile	Dimensions	Approx. quantity.
Sofa base and legs	Metalon 30x30 mm	140, 70, 34.5 cm	10 cuts approx.
Armchair structure	Metalon 30x30 mm	70.5, 70, 34.5 cm	8 cuts approx.
Tables (both)	Metalon 25x25 mm	85x60 and 45x45 cm	12 cuts approx.
Arms of wood	Pine or oak wood	4x6 cm (section) x 70 cm	2 pieces (optional for styling)

Additional material:

- Table surfaces (wood, MDF, glass or polished cement)
- Upholstered cushions or mattresses
- Fixing screws or direct welding
- Black anti-corrosion paint

4. Necessary tools

- MIG or inverter welder
 - Angle grinder (cutting and grinding)
 - Drill and bits for metal and wood
 - Tape measure, set square, level
 - Sander and spray paint
 - Wood saw (if wooden arms are included)
-

5. Cutting details

- **Main structures:** profile cuts for sides, crossbars, legs and supports.

- **Tables:**

- Coffee table: 2 frames of 85x60 cm and 4 legs of 43 cm
- Side table: 2 frames of 45x45 cm and 4 legs of 50 cm

- **Wooden arms (optional):** 2 pieces per seat (cut to size approx. 70 cm)

6. Difficulty level

Medium – Requires precision in measurements and symmetry for seating and solid structure.

7. Estimated manufacturing time

8 to 10 hours (including cushioning, painting, and complete set assembly)

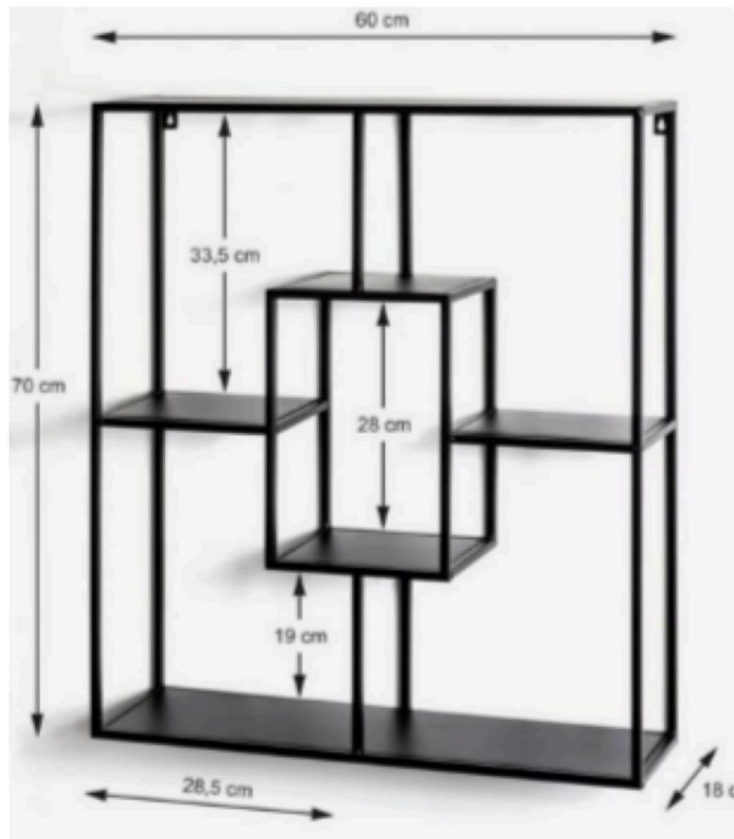
8. Assembly instructions

1. **Cut all the profiles** according to the design of the sofa, armchair and tables.
 2. **Weld side frames**, then join with lower and upper crossbars.
 3. **Add cross or perpendicular reinforcements** to support weight.
 4. **Fix wooden arms** to the side with screws or rivets (optional).
 5. **Sand all joints**, clean, and apply black paint.
 6. **Assemble table surfaces** (screw wood or fix glass).
 7. **Add cushions** to the seat and backrest, adjusting them with Velcro or straps if desired.
additional stability.
-

9. Additional tips

- If you use this set outdoors, consider applying **protective varnish** to the wooden arms and waterproof cushions.
- Legs may include **levelers or plastic caps** to prevent scratches.
- You can modify the structure to include storage space in the tables.

Proyecto 6: Estantería mural con cubo central suspendido



Project 6: Wall shelving with a suspended central cube

1. Description of the furniture

Modern metal wall shelving featuring a suspended central cube, ideal for decorating living rooms, offices, or studios. Its lightweight yet sturdy structure allows you to organize books, plants, or decorative objects on different levels in a geometric style.

2. General measures

- **Total width:** 60 cm
- **Total height:** 70 cm
- **Depth:** 18 cm
- **Left space height:** 33.5 cm
- **Bottom height under the bucket:** 19 cm
- **Height of the central cube:** 28 cm
- **Side space width (shelf depth):** 28.5 cm

3. Bill of materials (metal structure)

Material	Dimension is	Thickness (sheet)	Length (cm)	Approx. quantity.
Metalon square	20x20 mm Sheet Metal 18	70 cm	2 pieces (sides vertical)	
Metalon square	20x20 mm Sheet Metal 18	60 cm	1 piece (top horizontal)	
Metalon square	20x20 mm Sheet Metal 18	28.5 cm	2 pieces (inner bases)	
Metalon square	20x20 mm Sheet Metal 18	28 cm	4 pieces (structure of the central cube)	
Metalon square	20x20 mm Sheet Metal 18	18 cm	2 pieces (depth side)	

Additional material:

- 3 wooden or MDF shelves measuring 60 x 18 cm (or cut into segments)
- Screws for wall mounting or hidden bracket
- Matte black anti-corrosion paint

4. Necessary tools

- MIG or electric welder
 - Angle grinder (cutting disc)
 - Drill with bits for metal and wall
 - Hand sander or sandpaper
 - Level, set square and tape measure
 - Spray paint or brush
-

5. Cutting details (wooden shelves)

Shelf	Measurement (cm)
Lower shelf	60 x 18
	60 x 18
Intermediate shelf	
Top shelf	60 x 18

The central cube fits on the intermediate shelf, it can be fixed or loose for a visual floating effect.

6. Difficulty level

Medium – Accuracy required for center hub suspension.

7. Estimated manufacturing time

4 to 6 hours (including welding, painting, and wall mounting)

8. Assembly instructions

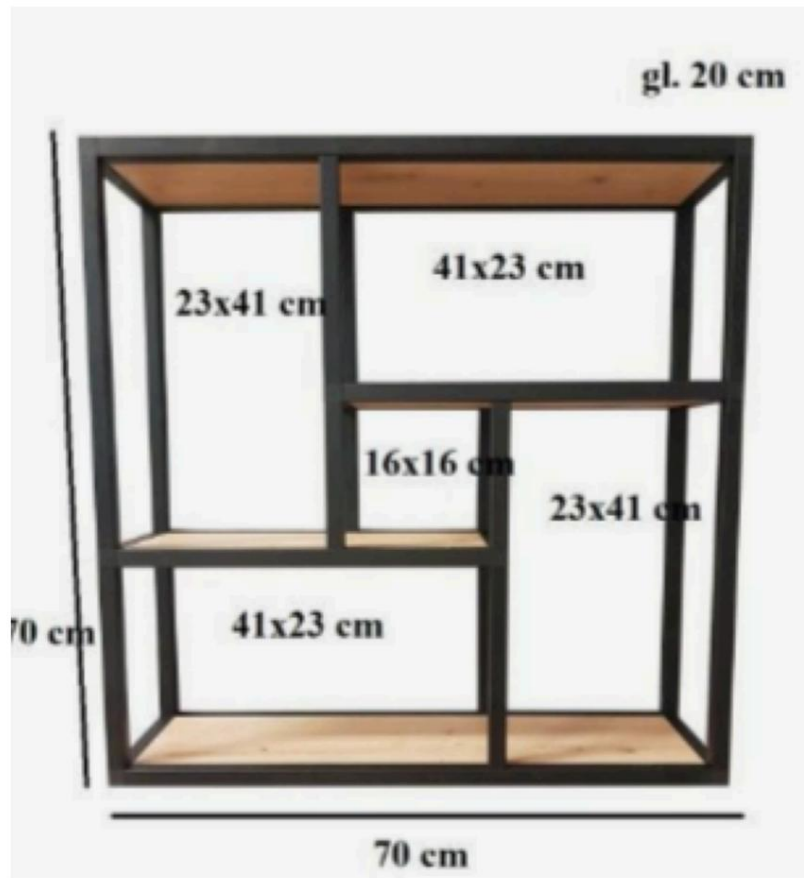
1. Cut all metal profiles according to the design.

2. **Assemble the external rectangular structure:** join the vertical sides with the top and bottom.
 3. **Install the interior shelves** at different heights by welding the brackets.
 4. **Build the central suspended cube** with 4 welded profiles forming a frame.
 5. **Fix the suspended cube** centered inside the module and joined at its ends sides.
 6. **Sand, clean and paint** the entire structure with a black finish.
 7. **Secure the frame to the wall** using hidden brackets or heavy-duty anchors.
-

9. Additional tips

- Make sure the center hub is level before welding.
- You can use **floating hardware** if you want a cleaner look without fixings visible.
- Ideal to combine with subtle LED lighting on the back if a more subtle look is desired. additional modern touch.

Proyecto 7: Estantería cuadrada con repisas modulares



Project 7: Square shelving with modular shelves

1. Description of the furniture

Square metal shelving unit with modular compartments of various sizes, ideal for organizing books, decorations, or office supplies. Its symmetrical and sturdy design, with shelves of varying heights and depths, combines practicality with an industrial aesthetic.

2. General measures

- **Total width:** 70 cm
- **Total height:** 70 cm
- **Depth:** 20 cm

3. Bill of materials (metal structure)

Material	Dimension is	Thickness (sheet metal)	Length (cm)	Approx. quantity.
Square metalon	20x20 mm Sheet Metal 18		70 cm	4 pieces (top, bottom and sides)
Metalon square	20x20 mm Sheet Metal 18		41 cm	4 pieces (internal dividers)
Square metalon	20x20 mm Sheet Metal 18		23 cm	4 pieces (horizontal depths)
Metalon square	20x20 mm Sheet Metal 18		16 cm	2 pieces (small center hub)

Additional material:

- 5 MDF or pine wood shelves (measurements: 41x23 cm, 23x41 cm and 16x16 cm)
- Screws or adhesive for fixing
- Matte black anti-corrosive paint

4. Necessary tools

- MIG or inverter welder
- Angle grinder with cutting and grinding discs
- Drill with bits for metal and wood
- Hand sander or sandpaper
- Tape measure, set square and level
- Spray paint or brush

5. Cutting details (wooden shelves)

Shelf	Measurement (cm)
Shelf 1 (top right)	41 x 23
Shelf 2 (top left)	23 x 41
Shelf 3 (bottom left)	41 x 23
Shelf 4 (bottom right)	23 x 41
Shelf 5 (center)	16 x 16

6. Difficulty level

Low – Ideal for those starting out in blacksmithing and wanting a structured and modular project.

7. Estimated manufacturing time

3 to 5 hours (includes assembly, painting, and shelf mounting)

8. Assembly instructions

1. **Cut all profiles** according to the design of the outer frame and internal divisions.
 2. **Assemble the main square frame** (70x70 cm).
 3. **Place the horizontal and vertical internal dividers** (23 cm, 41 cm, 16 cm) to form the modules.
 4. **Reinforce the central cube** by welding the four 16 cm pieces and fixing it to the frame.
 5. **Sand all welded joints**, clean well and apply black paint.
 6. **Place the wooden shelves** inside each compartment, securing them with discreet screws or strong adhesive.
-

9. Additional tips

- Ideal for installation with wall fixings if you want to anchor it to the wall.
- You can leave some spaces without shelves to place taller objects.
- If you want more robustness, you can use 25x25 mm profiles without losing aesthetics.

Proyecto 8: Librero vertical con módulos flotantes



Project 8: Vertical bookcase with floating modules

1. Description of the furniture

This tall bookcase combines floating compartments with structured modules that maximize vertical space. Designed for those seeking functionality and industrial aesthetics in a single piece. Ideal for offices, studios, or modern living rooms.

2. General measures

- **Total height:** 200 cm
 - **Total width:** 90 cm
 - **Total depth:** 40 cm
 - **Height of internal floating modules:** 21 cm, 44 cm, 28 cm
 - **Internal level heights:** 55 cm, 58 cm, 56 cm, 27 cm
-

3. Bill of materials (metal structure)

Material	Dimension is	Thickness (sheet)	Length (cm)	Approx. quantity.
Metalon square	25x25 mm Sheet 18		200 cm	2 pieces (vertical sides)
Metalon square	25x25 mm Sheet 18		90 cm	2 pieces (top and bottom)
Metalon square	25x25 mm Sheet 18		60 cm	2 pieces (top shelf and hanging module)
Metalon square	25x25 mm Sheet 18		56 cm	6 pieces (central modules and lower)
Metalon square	25x25 mm Sheet 18		30 cm	1 piece (low center support lower)
Metalon square	25x25 mm Sheet 18		40 cm	2 pieces (bottom depth)

Additional material:

- 4 wooden or MDF shelves with dimensions:
 - 60 x 40 cm
 - 56 x 40 cm (2 units)
 - 40 x 40 cm
- Black anti-corrosion paint

- Concealed screws or fasteners for wood
-

4. Necessary tools

- MIG or electric welder
 - Angle grinder
 - Drill with metal and wood bits
 - Set square, tape measure, level
 - Hand sander or sandpaper
 - Spray paint or brush
-

5. Cutting details (wooden shelves)

Shelf	Measurement (cm)
Top left	60 x 40
Floating central module	56 x 40
Right middle shelf	56 x 40
Lower shelf	40 x 40

6. Difficulty level

Advanced – Requires precision to ensure modules are level and symmetrical, in addition to stability throughout the structure.

7. Estimated manufacturing time

7 to 9 hours (including design, cutting, welding, painting, and assembly)

8. Assembly instructions

1. **Cut all profiles** according to the dimensions of the structure and modules.
 2. **Assemble the main frame** (200 cm high by 90 cm wide).
 3. **Add horizontal reinforcements** according to the internal divisions (levels 55, 58 and 56 cm).
 4. **Build the internal rectangular modules** and the floating module, leveling them correctly.
 5. **Weld the lower cross supports** and the 30 cm vertical reinforcement.
 6. **Sand and degrease the entire structure**, then apply black anti-corrosion paint.
 7. **Install the wooden shelves** on the welded frames, securing them from below or with hidden brackets.
-

9. Additional tips

- Add **rubber pads or levelers** to the base to prevent vibrations.
- You can include **LED lights** on the inner edges for a modern aesthetic.
- This design supports **modular customization**, adding or removing partitions without compromise the structure.

Proyecto 9: Silla industrial con respaldo de madera



Project 9: Industrial chair with wooden backrest

1. Description of the furniture

Industrial design chair with a sturdy metal frame, wooden seat, and decorative backrest. Perfect for dining rooms, desks, or loft-style bars. Its compact shape and modern aesthetic make it both functional and decorative.

2. General measures

- Total height: 95.5 cm
- Seat width: 41 cm
- Seat height: 47 cm

- **Seat depth:** 50 cm
 - **Wooden backrest height:** 20 cm
-

3. Bill of materials (metal structure)

Material	Dimension is	Thickness (sheet)	Length (cm)	Approx. quantity.
Metalon square	20x20 mm Sheet Metal	18	95.5 cm	2 pieces (rear sides)
Metalon square	20x20 mm Sheet Metal	18	47 cm	2 pieces (front legs)
Metalon square	20x20 mm Sheet Metal	18	41 cm	4 pieces (seat width and backup)
Metalon square	20x20 mm Sheet Metal	18	50 cm	2 pieces (depth of seat)
Metalon square	20x20 mm Sheet Metal	18	20 cm	1 piece (backrest reinforcement)

Additional material:

- 1 MDF or pine wood seat (41 x 50 cm)
 - 1 wooden backing board (41 x 20 cm)
 - Hidden screws or fasteners
 - Matte black anti-corrosive paint
-

4. Necessary tools

- MIG or electric welder
- Angle grinder
- Drill with bits for metal and wood
- Set square, tape measure and level

- Sander
- Spray paint or brush

5. Cutting details (wood)

Piece Measurement (cm)

Seat 41 x 50

Back 41 x 20

6. Difficulty level

Medium – Precision required to maintain symmetry and stability.

7. Estimated manufacturing time

4 to 5 hours (includes framing, painting, and wood assembly)

8. Assembly instructions

1. **Cut all metal profiles** according to measurements.
 2. **Assemble the seat base** by welding two 41 cm profiles and two 50 cm profiles.
 3. **Weld the front legs (47 cm)** to the front of the seat.
 4. **Join the 95.5 cm rear profiles** to the rear frame of the seat.
 5. **Weld the backrest reinforcements** including the lower crossbar (20 cm) and upper support (41 cm).
 6. **Sand and paint** the entire structure with a matte black finish.
 7. **Screw the backrest board** and the wooden seat onto the frame.
-

9. Additional tips

- You can **add a thin upholstered cushion** to the seat for greater comfort.
- Make sure the legs are **level** before fully welding.
- If it will be used outdoors, protect the wood with **water-resistant varnish**.

Proyecto 10: Estantería de tres niveles con estructura industrial



Project 10: Three-tier shelving with an industrial frame

1. Description of the furniture

Compact and elegant shelving unit with a black metal frame and three levels of wooden shelves. Ideal for use as a piece of auxiliary furniture in living rooms, kitchens, bathrooms, or entryways. Its open design allows for optimal organization without cluttering the space.

2. General measures

- **Total height:** 78 cm

- **Width:** 67 cm
 - **Depth:** 30 cm
 - **Height between shelves:** 30 cm (top), 32 cm (middle), 8 cm (bottom)
-

3. Bill of materials (metal structure)

Material	Dimension is	Thickness (sheet metal)	Length (cm)	Approx. quantity.
Square metalon	20x20 mm Sheet Metal 18		78 cm	4 pieces (vertical legs)
Metalon square	20x20 mm Sheet Metal 18		67 cm	3 pieces (top and crossbars)
Square metalon	20x20 mm Sheet Metal 18		30 cm	6 pieces (depths per level)

Additional material:

- 3 MDF or pine wood shelves measuring 67 x 30 cm
 - Screws for fixing or hidden brackets
 - Black anti-corrosion paint
-

4. Necessary tools

- MIG or electric welder
- Angle grinder
- Drill with metal and wood bits
- Hand sander or sandpaper
- Tape measure, set square and level
- Spray paint or brush

5. Cutting details (wooden shelves)

Shelf	Measurement (cm)
Top shelf	67 x 30
Medium shelf	67 x 30
Lower shelf	67 x 30

6. Difficulty level

Low – Ideal as a first industrial ironwork project.

7. Estimated manufacturing time

2.5 to 4 hours (including welding, painting, and assembly)

8. Assembly instructions

1. **Cut all metal profiles** according to measurements.
2. **Weld the upper frame** by joining 2 67 cm profiles with 2 30 cm profiles.
3. **Attach the vertical legs (78 cm)** to each corner of the top frame.
4. **Weld the 30 cm supports** on the intermediate and lower levels (at 30 cm and 62 cm cm from the ground respectively).
5. **Sand all joints**, clean with degreaser and paint.
6. **Mount the wooden shelves** on the metal frame, securing them with screws or strong glue.

9. Additional tips

- You can add **rubber feet** to the base to prevent scratches on floors.

- If you will be using it in humid environments, apply **protective varnish to the wood**.
- This structure can be easily adapted as a **TV table, shoe rack or mini bookseller**.

Proyecto 11: Silla industrial con brazos en ángulo y respaldo de rejilla



Project 11: Industrial chair with angled arms and mesh backrest

1. Description of the furniture

Chair with a sturdy metal frame, square seat, and a mesh backrest with a frame. Its straight front legs and angled back provide stability and a modern design.

It features armrests integrated into the upper sides, making it ideal for living rooms, offices, or industrial-style dining rooms.

2. General measures

- **Total height:** 82 cm
- **Seat height:** 45 cm

- **Armrest height:** 65 cm
- **Seat width:** 48 cm
- **Total depth:** 52 cm
- **Total width with arms:** 54 cm
- **Backrest height:** 33 cm

3. Bill of materials (metal structure)

Material	Dimension is	Thickness (sheet)	Length (cm)	Approx. quantity.
Metalon square	25x25 mm Sheet 18		82 cm	2 pieces (hind legs in angle)
Metalon square	25x25 mm Sheet 18		65 cm	2 pieces (front legs)
Metalon square	25x25 mm Sheet 18		54 cm	2 pieces (front and rear of the seat)
Metalon square	25x25 mm Sheet 18		48 cm	4 pieces (seat side and backrest frame)
Metalon square	25x25 mm Sheet 18		52 cm	2 pieces (horizontal arms superiors)

Additional material:

- 1 woven or slatted wooden base (48 x 48 cm)
- 1 piece of slatted backrest or rattan grid (48 x 33 cm approx.)
- Fixing screws or brackets
- Black anti-corrosion paint

4. Necessary tools

- MIG or electric welder

- Angle grinder
- Drill with metal and wood bits
- Hand sander or sandpaper
- Tape measure, set square and level
- Paint gun or brush

5. Cutting details (wood/lattice)

Part	Measurement (cm)
Base seat	48 x 48
Back	48 x 33

6. Difficulty level

Advanced – Due to its angled structure and connection of backrest with arms, it requires experience in alignment and symmetry.

7. Estimated manufacturing time

5 to 7 hours (including painting and grid back assembly)

8. Assembly instructions

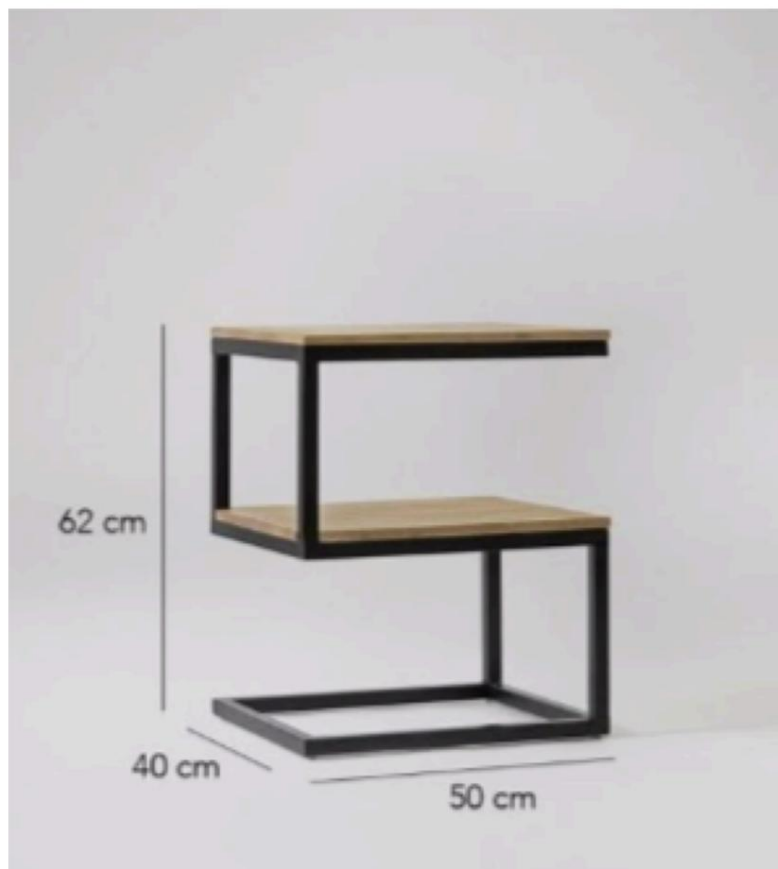
1. **Cut all metal profiles** according to the indicated measurements.
2. **Form the base seat** by welding 2 profiles of 54 cm and 2 of 48 cm.
3. **Weld the front legs (65 cm)** perpendicular to the frame.
4. **Place the rear legs (82 cm)** at an angle and weld them to the frame.
5. **Add the upper armrests (52 cm)** between the top of the legs rear and front.

6. **Build the backrest frame** with 48x33 cm profiles and weld it together. arms.
 7. **Sand, clean and paint** the entire structure.
 8. **Secure the seat base** and rear grille inside the backrest frame.
-

9. Additional tips

- You can use **wicker, rattan or braided PVC** for the backrest.
- Make sure the arms are **completely level** before welding.
- If you want greater comfort, you can upholster the seat base.

Proyecto 12: Mesa auxiliar en Z con estructura metálica





Project 12: Z-shaped side table with metal frame

1. Description of the furniture

A Z-shaped side or auxiliary table, ideal for living rooms, bedrooms, or reading spaces. Its matte black metal frame and wooden shelves give it a modern industrial style. In addition to being decorative, it's also functional for displaying lamps, books, or drinks.

2. General measures

- **Total height:** 62 cm
- **Width:** 50 cm
- **Depth:** 40 cm

3. Bill of materials (metal structure)

Material	Dimension is	Thickness (sheet metal)	Length (cm)	Approx. quantity.
Square metalon	25x25 mm Sheet 18		50 cm	4 pieces (front and back of each level)

Metalon square	25x25 mm Sheet 18	40 cm	6 pieces (sides and depth of bases)
Square metalon	25x25 mm Sheet 18	22 cm	2 pieces (central vertical joint type Z)

Additional material:

- 2 MDF or pine wood shelves (50 x 40 cm)
- Fixing screws or invisible brackets
- Black anti-corrosion paint

4. Necessary tools

- MIG or inverter welder
- Angle grinder (cutting and grinding discs)
- Drill with metal/wood bits
- Hand sander or sandpaper
- Tape measure, level and set square
- Spray paint or brush

5. Cutting details (wooden shelves)

Shelf	Measurement (cm)
Upper level	50 x 40
Lower level	50 x 40

6. Difficulty level

Low – Ideal as a decorative or functional project for interiors.

7. Estimated manufacturing time

2.5 to 3.5 hours (includes framing, painting, and wood fixing)

8. Assembly instructions

1. **Cut all metal profiles** according to measurements.
 2. **Assemble the lower frame** (50 x 40 cm) as the base of the furniture.
 3. **Build the top frame** with the same dimensions.
 4. **Weld the central vertical joints** (approx. 22 cm) offset to form the "Z".
 5. **Check that the surfaces are level**, weld completely and reinforce structural points.
 6. **Sand the joints and apply black anti-corrosion paint.**
 7. **Place the wooden shelves** on both levels and secure them from below with hidden screws or brackets.
-

9. Additional tips

- This design can **function as a nightstand** or entry table if adapted the high one.
- You can add **small wheels** if mobility is desired.
- For a more modern style, you can use **melamine, marble or glass boards tempered.**

Proyecto 13: Estantería alta de cinco niveles con estructura abierta



Project 13: Five-tier open-frame tall shelving

1. Description of the furniture

Industrial-style open vertical shelving unit with a metal frame and five wooden shelves. Its design maximizes storage height, making it ideal for books, plants, decorative objects, or even as a room divider.

2. General measures

- **Total height:** 160 cm
 - **Total width:** 120 cm
 - **Depth:** 34 cm
 - **Number of shelves:** 5 levels
-

3. Bill of materials (metal structure)

Material	Dimension is	Thickness (sheet metal)	Length (cm)	Approx. quantity.
Square metalon	30x30 mm Sheet 18		160 cm	4 pieces (vertical legs)
Metalon square	30x30 mm Sheet 18		34 cm	10 pieces (horizontal connectors, 2 per level)

Additional material:

- 5 MDF, pine or melamine wood shelves measuring 120 x 34 cm
- Screws for fixing or hidden brackets
- Black anti-corrosion paint

4. Necessary tools

- MIG or inverter welder
- Angle grinder (cutting and grinding discs)
- Drill with bits for metal and wood
- Square, level, tape measure
- Hand sander or sandpaper
- Spray paint or brush

5. Cutting details (wooden shelves)

Shelf	Measurement (cm)
All the shelves	120 x 34

6. Difficulty level

Medium – Requires leveling precision to ensure shelves are aligned and secure.

7. Estimated manufacturing time

4 to 6 hours (includes metalwork, painting, and shelf assembly)

8. Assembly instructions

1. **Cut the 160 cm profiles** (4 pieces) for the vertical columns.
 2. **Cut 10 profiles of 34 cm** to use as horizontal connectors on both sides from each shelf.
 3. **Weld two 34 cm profiles** between two columns at the same height to form a level.
 4. **Repeat the operation for the five levels**, taking care to leave proportional distances between each one.
 5. **Sand the welded joints**, clean with degreaser and paint with paint. black anticorrosive.
 6. **Place the wooden shelves** against the frames or secure them from below with screws or invisible brackets.
-

9. Additional tips

- You can add **rear X-cross braces** for added stability if use is demanding.
- Ideal for use in offices, living rooms or kitchens as a display unit or library.
- For an extra decorative touch, you can use **recycled or fire-treated wood**.

Proyecto 14: Cama alta con escaleras y estructura metálica multifuncional



Project 14: Loft bed with stairs and metal frame multifunctional

1. Description of the furniture

A loft bed with side access via stairs and a reinforced metal frame, ideal for optimizing space in small rooms. The lower level is free and can be used as a desk, play area, or storage area. The modular side ladder also functions as a shelf or bookcase.

2. General measures

- **Total height:** 174 cm
- **Total width (with stairs):** 290 cm
- **Bed only width:** 240 cm
- **Depth (bottom):** 133 cm
- **Free height under bed:** 137 cm
- **Steps:** 6 pieces, each approx. 21.5 cm high

3. Bill of materials (metal structure)

Material	Dimension is	Thickness (sheet metal)	Estimated length	Approx. quantity.
Rectangular metalon	40x40 mm Sheet	16 Variable	(total 25–30 m)	Main structure and railings
Metalon square	30x30 mm Sheet	18 Variable	(12–15 m)	Reinforcements and stair structure
Metal plate or angle	—	—	—	Support for safety steps and sides

Additional material:

- 6 solid wood steps (width 51.4 cm)
- 1 wooden bed base measuring 240 x 133 cm
- Upper guardrails (can be made of metal or wood)
- Screws, anchors and reinforcement hardware
- Black anti-corrosion paint

4. Necessary tools

- MIG or inverter welder
- Angle grinder (cutting and grinding discs)
- Drill with bits for metal and wood
- Set square, tape measure, level
- Hand sander or sandpaper
- Paint gun or brush

5. Cutting details (wood)

Part	Measurement (cm)
Bed base	240 x 133
Stair steps	51.4 x 25 approx. (x6)
Optional – handrails	240 x 20 (front and side)

6. Difficulty level

Advanced – Large-scale structure, requires experience in welding and secure structural assembly.

7. Estimated manufacturing time

12 to 16 hours (including framing, painting, wood installation, and leveling)

8. Assembly instructions

1. **Cut all metal profiles** according to the measurements of the frame, railings and staircase.
 2. **Assemble the base structure of the bed**, joining the sides, rails and supports transversals.
 3. **Weld the vertical legs (174 cm)** and fix them to the base frame.
 4. **Assemble the bookcase-type side staircase**, ensuring that the steps are leveled and well welded.
 5. **Add top rails** on the free sides for safety.
 6. **Sand and paint** the entire metal structure with a matte black anti-corrosive finish.
 7. **Install the wooden base for the mattress** and screw each rung onto the metal frame of the ladder.
-

9. Additional tips

- Secure the structure **to the wall or floor** if it is to be used by children or older people. security.
- You can convert the lower part into a fixed desk or play area.
- If disassemblability is desired, consider using **bolt-on angles** instead of fixed welding on some parts.

Proyecto 15: Set de sala exterior industrial (sofá, sillones y mesa de centro)



Project 15: Industrial outdoor living room set (sofa, armchairs, and coffee table)

1. Description of the furniture

This industrial-style living room set includes a two-seater sofa, two armchairs, and a rectangular coffee table. With a black metal frame and sturdy cushions, it's perfect for covered outdoor spaces, terraces, patios, or modern social areas.

2. General measurements (converted from inches to centimeters)

Two-seater sofa

- Width: 48.4 in ÷ 123 cm

- Depth: 26.4 in ÷ 67 cm
- Total height: 25.2 in ÷ 64 cm
- Seat height: 13.8 in ÷ 35 cm

Individual armchairs

- Width: 27.2 in ÷ 69 cm
- Depth: 26.4 in ÷ 67 cm
- Total height: 25.2 in ÷ 64 cm

Coffee table

- Length: 40.2 in ÷ 102 cm
- Width: 21.5 in ÷ 55 cm
- Height: 13.8 in ÷ 35 cm

3. Bill of materials (metal structure)

Material	Dimensions s	Thickness (sheet metal)	Use
Metalon square	30x30 mm Sheet 18		Structure of sofas and table
Square metalon	25x25 mm Sheet 18		Internal reinforcements

Additional material:

- 4 large cushions (seat)
- 4 medium cushions (backrest)
- 1 coffee table top (wood, slats or plastic composite)
- Screws, rubber plugs or ferrules

- Black anti-corrosion paint
-

4. Necessary tools

- MIG or inverter welder
 - Angle grinder (cutting and grinding)
 - Drill with metal/wood bits
 - Sander or sandpaper
 - Set square, tape measure and level
 - Spray paint or brush
-

5. Cutting details (structure per piece)

Two-seater sofa:

- 2 side frames: 64 x 67 cm
- 2 crossbars: 123 cm (top and bottom)
- Reinforcements for seat base and backrest

Armchairs (x2):

- 2 side frames per chair: 64 x 67 cm
- 2 crossbars per chair: 69 cm
- Internal reinforcements

Coffee table:

- 2 sides: 35 x 55 cm
- 2 side members: 102 cm

- 4 wooden slats for surface

6. Difficulty level

Medium – Requires precision to align frames and reinforcements.

7. Estimated manufacturing time

8 to 10 hours (the entire set, including painting and cushion assembly)

8. Assembly instructions

1. **Cut and weld side frames** for sofas and armchairs.
 2. **Join with front and rear crossbars**, forming each rectangular structure.
 3. **Add internal supports** to the base to support cushions or rigid bases.
 4. **Sand and paint** the pieces with black anti-corrosion paint.
 5. **Cut and assemble the coffee table** using the same process.
 6. **Place the wooden surfaces** on the table and the cushions on the seats.
-

9. Additional tips

- Use **waterproof or outdoor cushions** if you will be using it outdoors.
- You can add **non-slip rubber** to the legs for greater stability.
- The design can be scaled to three seats or include more modules (side tables, chaise lounge).